

SUPERVISED MACHINE LEARNING
Support Vector Regressor

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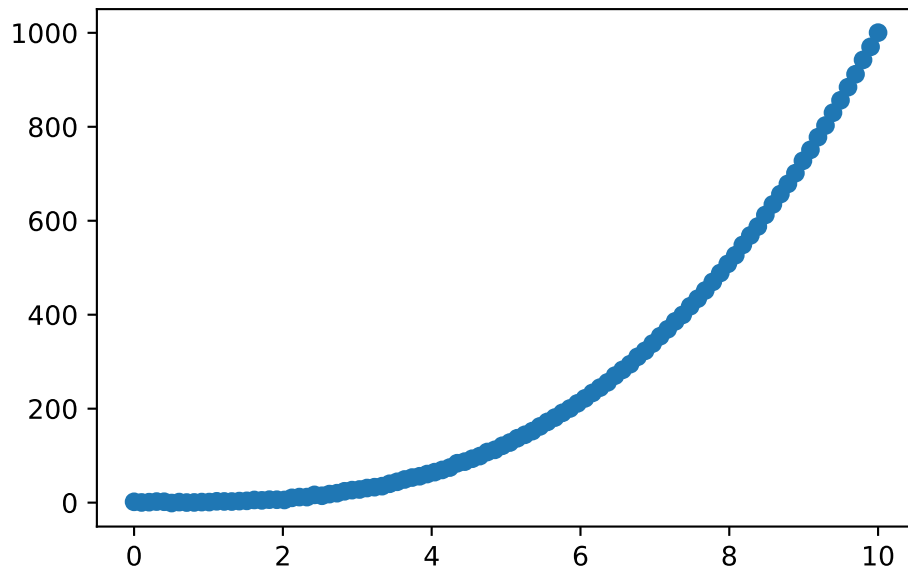
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1) Directly created by sklearn

```
1 import numpy as np
2 import matplotlib.pyplot as plt
3 import pandas as pd
4
5 from sklearn.linear_model import LinearRegression
6 from sklearn.svm import SVR
7 from sklearn.datasets import make_regression
8 from sklearn.metrics import mean_squared_error, r2_score
9
10 np.random.seed(0)
11 m = 100
12 X = np.linspace(0,10,m).reshape(m,1)
13 y = X**3 + np.random.randn(m,1)
14 X.shape
15 y.shape
16
17 # Initial data plot
18 plt.close()
19 plt.scatter(X,y)
20 plt.show()
21
22 # SVR of SVM
23 svr = SVR(C=100)
24 y = y.ravel()
25 svr.fit(X, y)
26 y_pred = svr.predict(X)
27
28 R2 = r2_score(y, y_pred)
29 rmse = mean_squared_error(y, y_pred)
30
31 print(f"r2_score = {R2} and .score = {svr.score(X, y)}")
32 print(f"RMSE = {rmse}")
```

```
24  
25 plt.close()  
26 plt.scatter(X, y)  
27 plt.plot(X, y_pred, c = 'r')  
28 plt.show()
```



$r^2_{\text{score}} = 0.9912970268952965$ and $\text{.score} = 0.9912970268952965$
RMSE = 720.2209881096592

